

ANDIR, China

WMO: 518480

Lat: 37.933N

Lon: 83.650E

Elev: 1264

StdP: 87.03

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-15.9	-23.5	0.5	-6.5	-22.2	0.6	-7.3	5.4	-2.4	4.3	-0.1	0.4	200	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	37.2	17.1	35.9	16.6	34.5	16.3	19.0	31.5	18.2	30.9	17.6	30.4	1.9	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	12.9	22.1	14.5	12.0	21.9	13.4	11.2	22.1	59.8	31.4	56.9	31.2	54.7	30.5	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.4	4.7	4.0	-20.2	40.0	1.4	1.5	-21.3	41.1	-22.1	42.0	-22.9	42.8	-24.0	43.9
			-20.5	21.9	1.4	1.4	-21.5	22.9	-22.3	23.8	-23.1	24.6	-24.1	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-7.4	-1.5	6.8	15.0	20.1	24.4	26.1	24.7	19.6	10.6	2.3	-5.4		
	DBStd	12.10	3.09	3.19	3.51	4.01	3.51	2.52	2.49	2.70	2.96	3.46	4.11	3.18		
	HDD10.0	1718	539	323	109	4	1	0	0	0	0	34	232	477		
	HDD18.3	3324	797	556	358	114	23	1	0	1	20	239	481	736		
	CDD10.0	2211	0	0	9	156	313	433	500	456	289	54	1	0		
	CDD18.3	773	0	0	0	14	77	183	242	198	58	0	0	0		
	CDH23.3	11079	0	0	0	341	1191	2424	3290	2734	1042	57	0	0		
	CDH26.7	5553	0	0	0	106	473	1232	1839	1497	404	3	0	0		
(14) Wind	WSAvg	1.5	1.1	1.3	1.9	2.0	1.8	1.7	1.5	1.6	1.4	1.2	1.1	0.9		
(15) Precipitation	PrecAvg	20	0	1	0	1	2	5	8	2	1	0	0	0		
	PrecMax	42	3	5	7	20	24	19	31	8	6	4	1	2		
	PrecMin	4	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	11	1	1	1	4	5	6	8	3	1	1	0	0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	12.5	21.7	32.6	35.0	37.4	39.6	38.6	35.1	26.7	19.7	8.4		
		MCWB	-1.1	2.9	7.0	13.1	14.9	17.3	18.5	17.2	15.5	11.1	7.0	1.3		
	2%	DB	3.2	10.1	19.8	29.6	33.0	35.6	37.5	36.8	33.0	25.1	16.6	5.2		
		MCWB	-2.4	2.0	6.4	12.0	14.2	16.2	17.5	16.8	15.0	10.2	5.5	-0.7		
	5%	DB	1.2	8.0	17.6	27.3	31.2	34.3	36.1	35.5	30.7	23.0	14.0	3.0		
		MCWB	-3.5	0.7	5.5	11.2	13.4	16.0	17.2	16.3	14.0	9.6	4.4	-2.1		
	10%	DB	-0.6	6.1	15.6	24.8	29.5	32.9	34.5	33.9	29.0	20.5	11.2	1.2		
		MCWB	-4.6	-0.5	4.6	10.0	12.9	15.6	17.0	16.1	13.3	8.3	3.1	-3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	3.5	7.6	14.3	16.8	19.6	20.7	19.7	17.1	11.9	7.2	1.5		
		MCDB	5.3	11.5	20.4	29.4	28.1	32.3	32.6	30.6	31.3	24.6	19.0	7.8		
	2%	WB	-2.3	2.2	6.6	12.5	15.2	18.3	19.2	18.4	15.8	10.7	5.7	-0.4		
		MCDB	2.9	9.8	18.4	28.6	30.0	31.1	32.0	30.7	29.6	23.6	16.0	4.1		
	5%	WB	-3.4	0.8	5.7	11.5	14.4	17.4	18.5	17.7	14.9	9.8	4.5	-1.7		
		MCDB	1.1	7.6	16.8	26.5	28.8	30.5	31.3	30.7	27.6	22.4	13.6	2.5		
	10%	WB	-4.6	-0.4	4.8	10.3	13.6	16.6	17.9	17.1	14.1	8.6	3.3	-2.9		
		MCDB	-0.6	5.7	14.9	23.8	27.5	29.5	30.6	30.6	26.1	19.9	11.0	1.3		
(35) Mean Daily Temperature Range	MDBR		13.8	14.7	14.5	15.3	15.5	14.7	14.7	16.2	18.0	18.5	16.8	13.8		
	5% DB	MCDBR	16.0	16.6	18.5	19.1	19.4	18.7	18.8	20.2	21.0	21.5	20.2	16.0		
		MCWBR	11.5	10.4	9.3	8.3	7.1	6.2	6.1	6.5	7.9	9.5	11.1	11.0		
	5% WB	MCDBR	15.5	15.8	17.1	17.9	16.4	14.4	14.2	16.2	18.2	20.0	19.8	15.0		
		MCWBR	11.2	10.0	8.7	8.0	7.0	5.6	5.5	5.7	7.2	8.8	11.0	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.342	0.408	0.511	0.601	0.635	0.654	0.632	0.632	0.585	0.451	0.380	0.349		
	taud		2.062	1.898	1.662	1.503	1.497	1.513	1.578	1.566	1.613	1.820	1.975	2.041		
	Ebn at Noon		805	789	739	701	689	676	687	671	668	729	745	761		
	Edh at Noon		121	163	229	285	291	287	267	264	238	174	131	115		
(44) All-Sky Solar Radiation	RadAvg		2.86	3.79	5.18	6.18	6.92	6.97	6.76	6.26	5.62	4.66	3.33	2.65		
	RadStd		0.32	0.21	0.24	0.28	0.28	0.28	0.33	0.40	0.18	0.11	0.15	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		-0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-139	-117

Nomenclature: See separate page